



**MinersBuddy**

EXAM PREPARATION

COMPLETE EXAM SYLLABUS

# GATE MINING ENGINEERING (MN)

GATE 2027 | Organising Institute: IIT Madras

Official syllabus, restructured for study

**6**

Sections

**110**

Topics

**100**

Marks

# Syllabus at a Glance

GATE 2027 Mining Engineering (MN): Engineering Mathematics plus five core mining sections. Topics follow the official IIT Madras syllabus document.

| NO.   | SECTION                                                       | TOPICS |
|-------|---------------------------------------------------------------|--------|
| 1     | Engineering Mathematics                                       | 27     |
| 2     | Mining Geology, Mine Development and Surveying                | 21     |
| 3     | Geomechanics and Ground Control                               | 16     |
| 4     | Mining Methods and Machinery                                  | 7      |
| 5     | Surface Environment, Mine Ventilation and Underground Hazards | 20     |
| 6     | Mineral Economics, Mine Planning, Systems Engineering         | 19     |
| Total |                                                               | 110    |

| EXAM PATTERN      | DETAILS                                        |
|-------------------|------------------------------------------------|
| Mode              | Computer-Based Test (CBT), English             |
| Duration          | 3 hours (180 minutes), no sectional time limit |
| Questions / Marks | 65 questions, 100 marks                        |

Note: General Aptitude (common to all GATE papers) is not part of the MN syllabus document, so it is not listed here. Check the official GATE 2027 General Aptitude syllabus and the latest exam pattern on [gate2027.iitm.ac.in](http://gate2027.iitm.ac.in) before you plan.

**1****Engineering Mathematics****Linear Algebra**

- Matrices and determinants
- Inverse and rank of matrix
- Systems of linear equations
- Eigen values and eigen vectors

**Calculus**

- Limit, continuity and differentiability
- Partial derivatives
- Mean value theorems
- Indeterminate forms and L' Hospital's rule
- Maxima and minima
- Taylor's theorem
- Sequences and series
- Fourier series

**Vector Calculus**

- Gradient, divergence and curl
- Line, surface and volume integrals
- Stokes, Gauss and Green's theorems

**Differential Equations**

- Linear and non-linear first order ODEs
- Higher order linear ODEs with constant coefficients
- Cauchy's and Euler's equations

**Probability and Statistics**

- Probability: joint, conditional, Bayes' theory
- Descriptive statistics
- Sampling theory and hypothesis testing
- Binomial, Poisson, exponential and normal distributions
- Pearson correlation and linear regression

**Numerical Methods**

- Solutions of linear algebraic equations
- Interpolation
- Integration: trapezoidal and Simpson's rule
- Single and multi-step methods for differential equations

**2****Mining Geology, Mine Development and Surveying****Mining Geology**

- Minerals, rocks and their origin
- Classification
- Ore genesis
- Structural geology

### Mine Development

- Methods of access to deposits
- Underground drivages
- Drilling methods and machines
- Explosives and initiation system
- Blasting tools and blast design practices
- Ground vibration

### Mine Surveying

- Levels and levelling
- Tacheometry
- Triangulation
- Errors and adjustments
- Correlation
- Underground surveying
- Curves and contouring
- Photogrammetry
- Theodolite, EDM, Total Station, GPS
- Remote sensing
- Astronomical survey

## 3

## Geomechanics and Ground Control

### Engineering Mechanics

- Equivalent force systems
- Equations of equilibrium
- Two dimensional frames and trusses
- Friction forces
- Particle kinematics and dynamics
- Beam analysis

### Geomechanics

- Geo-technical properties of rocks
- Rock mass classification
- Instrumentation and in-situ stress measurement techniques
- Theories of rock failure
- Stress distribution around mine openings
- Subsidence
- Slope and dump stability

### Ground Control

- Design of pillars and supporting systems
- Mine filling
- Strata control

## 4

## Mining Methods and Machinery

### Mining Methods

- Surface mining: layout and haul road development, loading, transportation and mechanization, continuous mining systems; highwall mining
- Underground coal mining: bord and pillar systems, longwall mining, thick seam mining methods
- Underground metal mining: open, supported and caving stoping methods, stope mechanization, ore handling systems

### Mining Machinery

- Generation and transmission of mechanical, hydraulic and pneumatic power
- Materials handling: wire ropes, haulages, conveyors, face and development machineries, hoisting systems, pumps
- Comminution methods and machinery
- Rock-tool interaction and mechanical cutting

**5****Surface Environment, Mine Ventilation and Underground Hazards****Surface Environment**

- Air, water and soil pollution: standards of quality, causes and dispersion of contamination and control
- Noise pollution and control
- Land reclamation
- EIA

**Mine Ventilation**

- Underground atmosphere
- Heat load sources and thermal environment, air cooling
- Mechanics of airflow, distribution, natural and mechanical ventilation
- Mine fans and usage
- Auxiliary ventilation
- Ventilation survey and planning
- Ventilation networks

**Underground Hazards**

- Mine gases
- Methane drainage
- Hazards from fires, explosions, dust and inundation
- Rescue apparatus and practices
- Safety management plan
- Safety data analytics
- Mine illumination
- Mine legislation
- Occupational health and safety

**6****Mineral Economics, Mine Planning, Systems Engineering****Mineral Economics**

- Mineral resource classification
- Discounted cash flow analysis
- Mine valuation
- Mineral taxation

**Mine Planning**

- Sampling methods, practices and interpretation
- Reserve estimation techniques: basics of geostatistics and quality control
- Optimization of facility location
- Mine planning and its components
- Determination of mine size and mine life
- Ultimate pit design
- Grade-tonnage relationship, ore dilution, recovery and cut-off grade
- Equipment selection

**Systems Engineering**

- Concepts of reliability
- Reliability of simple systems
- Maintainability and availability
- Linear programming, transportation and assignment problems
- Network analysis
- Inventory models
- Queuing theory

## Reference

| REFERENCE                                 | SOURCE                                                                       |
|-------------------------------------------|------------------------------------------------------------------------------|
| MN Mining Engineering Syllabus, GATE 2027 | IIT Madras ( <a href="https://gate2027.iitm.ac.in">gate2027.iitm.ac.in</a> ) |